Investigating the impact of spacecraft's configuration to detect current in space, in the context of the NASA HelioSwarm mission.

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Master 2 SSA- Fundamental Physics and Application (2025-2026)
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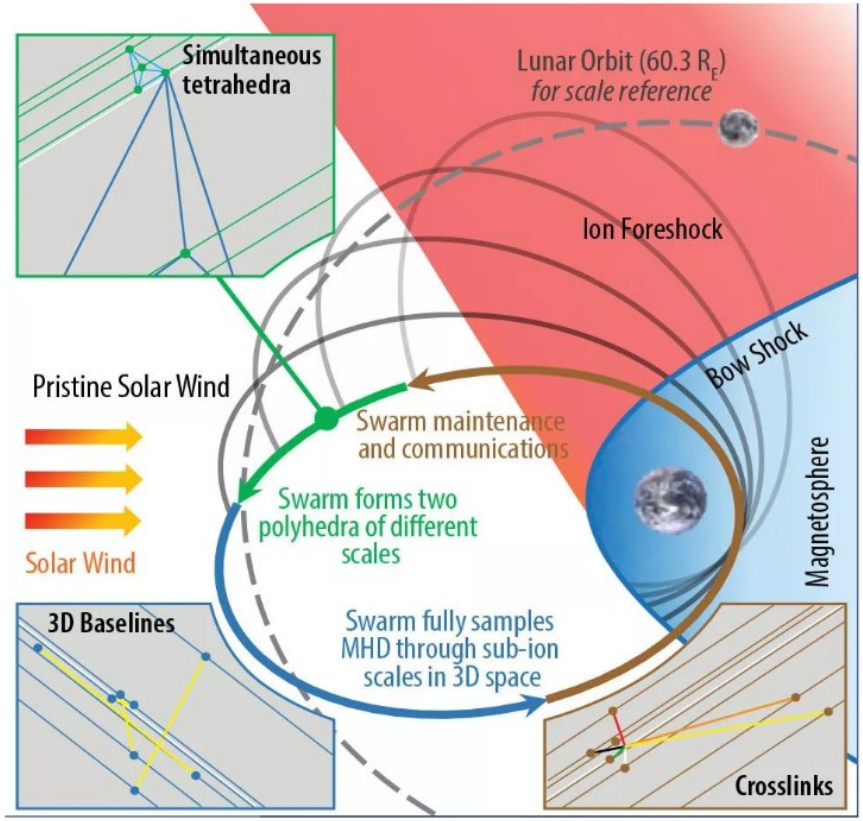
(Under the supervision of Prof. Matthieu Kretzschmar & Mehul Chakraborty)

HelioSwarm mission

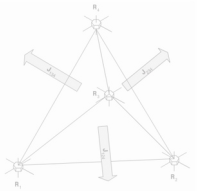
HelioSwarm's first-ever simultaneous multipoint, multiscale measurements disentangle spatial and temporal variations in solar wind plasmas that connect MHD scale turbulence with sub-ion scale heating.



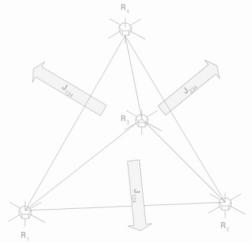
Source:https://eos.unh.edu/helioswarm/mission

The challenge:

We have four spacecraft flying in formation. Each measures the local magnetic field B. From these four vector measurements, we must reconstruct the three-dimensional current density **J** flowing through the tetrahedron.

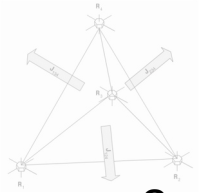


Two complementary approaches exist:

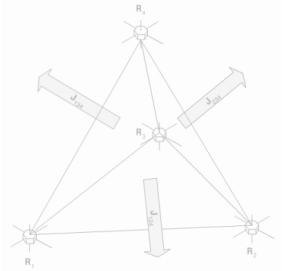


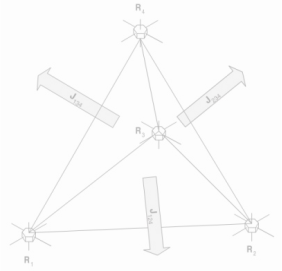
Curlometer – Direct discretization of Ampère's law (established, but geometry-limited)(*Curlometer history: Born ~1998 for ESA's Cluster mission (4 SC, launched 2000) —Malcolm Dunlop (UK) pioneered it to compute ∇×B → J from tetrahedron B-gradients, revolutionizing in-situ currents without models.*)

J-filtering – Grid-based current inversion or Biot-Savart grid optimization(recent development from LPC2E)



Physical Foundation – Ampère's Law

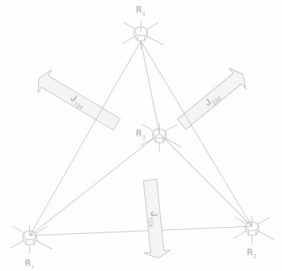




$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}$$

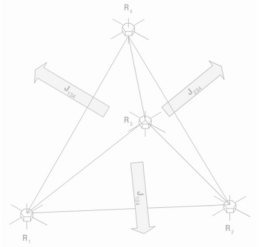
The curl of the magnetic field directly gives the current density.

The question is: how do we estimate this curl from four discrete measurements?



The Curlometer Method – Mathematical Framework

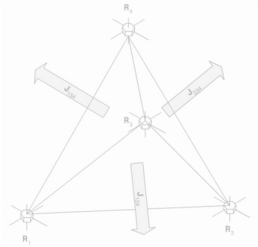
- The curlometer is the classical approach, widely used in Cluster and MMS data analysis. It makes a critical assumption: linear magnetic field variation across the tetrahedron.



Step 1: Linear Field Approximation:

$$\mathbf{B}(\mathbf{r}) \approx \mathbf{B}_c + \nabla\mathbf{B} \cdot (\mathbf{r} - \mathbf{r}_c)$$

$\mathbf{B}_c$ is the average field: $\mathbf{B}_\mathbf{C} = \frac{1}{4}\sum_{\alpha=1}^{4} \mathbf{B}_\alpha$



Step 2: Volumetric Tensor Construction:

From the four spacecraft positions we construct the volumetric tensor :

$$R_{jk} = \frac{1}{4}\sum_{\alpha=1}^{4}(r_{\alpha,j} - r_{c,j})(r_{\alpha,k} - r_{c,k})$$

This symmetric **3X3** matrix encodes the tetrahedron's shape and size. Its eigenvalues determine geometric quality.

The Curlometer Method – Mathematical Framework

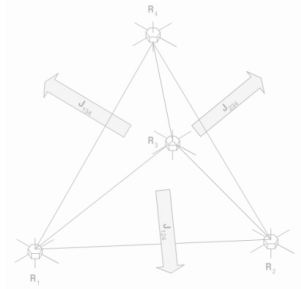
Step 3: Gradient Estimation via Least-Squares:

The magnetic field gradient is obtained by inverting R: $$\nabla \mathbf{B} = \frac{1}{4}\left(\sum_{\alpha=1}^{4}(\mathbf{B}_\alpha - \mathbf{B}_c)\otimes(\mathbf{r}_\alpha - \mathbf{r}_c)\right)\mathbf{R}^{-1}$$

This inversion is the Achilles' heel of the method. If ***R*** is ill-conditioned (tetrahedron nearly planar or elongated), small measurement errors get catastrophically amplified.

Step 4: Current Calculation:

Finally, the current density is: $$\mathbf{J}_{\mathrm{curl}} = \frac{1}{\mu_0}\begin{pmatrix}\partial_y B_z - \partial_z B_y \\ \partial_z B_x - \partial_x B_z \\ \partial_x B_y - \partial_y B_x\end{pmatrix}$$



Limitation: This entire procedure assumes constant current density across the constellation volume.

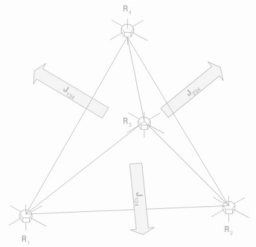
It works only for large-scale currents much larger than inter-spacecraft distances.

J-Filtering – A Fundamentally Different Philosophy

- **J-filtering**, developed by *Prof. Matthieu Kretzschmar, Mehul Chakraborty and Prof.J.L. Pinçon* at LPC2E, takes a **forward-modeling approach** inspired by k-filtering techniques.

- **Core idea:** Instead of inverting measurements to get current, search through physically plausible current configurations and find the one that best explains the observed magnetic fields.

J-Filtering – A Fundamentally Different Philosophy

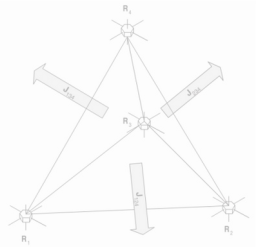


Forward Model – Biot-Savart Calculation

$$\mathbf{B}(\mathbf{r}_{sc}) = \frac{\mu_0}{2}\hat{\mathbf{j}} \times \frac{\mathbf{r}_{sc} - \mathbf{r}_0}{|\mathbf{r}_{sc} - \mathbf{r}_0|^3}$$

- Our forward model implements the Biot-Savart law for uniform currents over spacecraft scales. Our code computes cross-product components directly.

Preprocessing – Remove Global Fields



$$\mathbf{B}(t) = \begin{pmatrix} \mathbf{B}(\mathbf{r}_1, t) - \langle \mathbf{B}(t) \rangle \\ \mathbf{B}(\mathbf{r}_2, t) - \langle \mathbf{B}(t) \rangle \\ \mathbf{B}(\mathbf{r}_3, t) - \langle \mathbf{B}(t) \rangle \\ \mathbf{B}(\mathbf{r}_4, t) - \langle \mathbf{B}(t) \rangle \end{pmatrix}$$

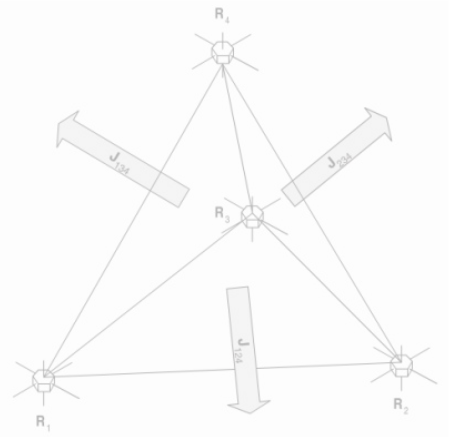
This removes large-scale background fields not produced by local currents.

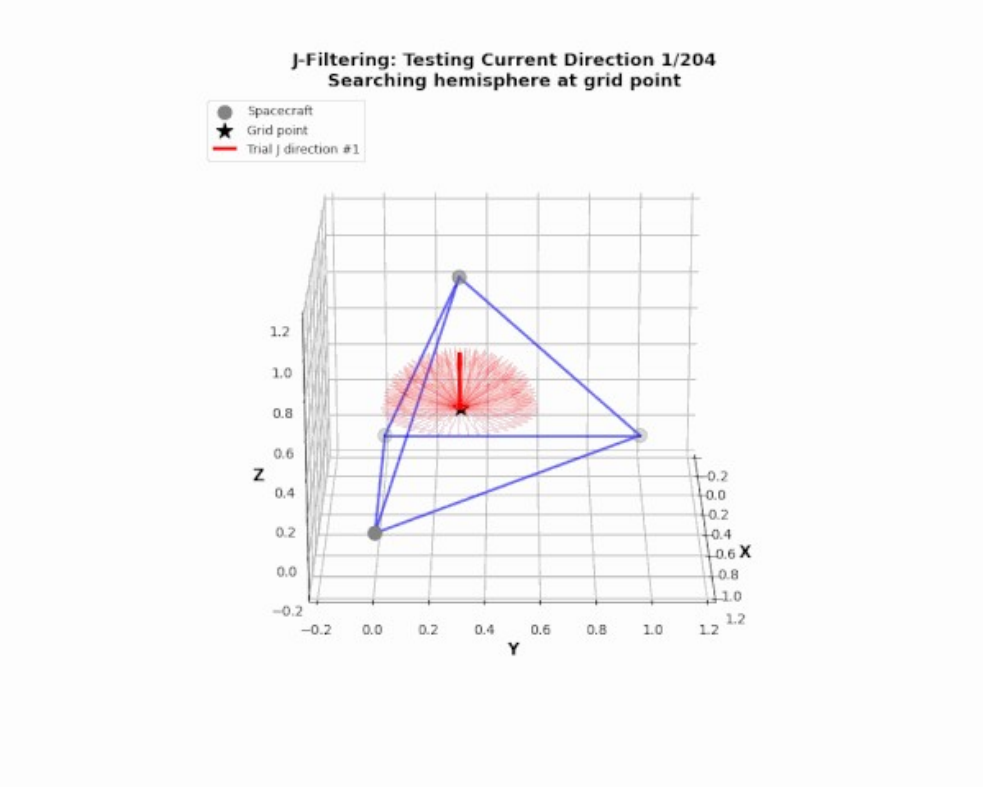
where $\langle \mathbf{B}(t) \rangle$ is the constellation-averaged field.

J-Filtering – A Fundamentally Different Philosophy

Define Search Space:

Create a 3D grid covering the tetrahedron volume. At each grid point $\mathbf{r}_i$, test ~ 100–200 trial current directions $\hat{\mathbf{u}}_j$ forming a hemisphere.



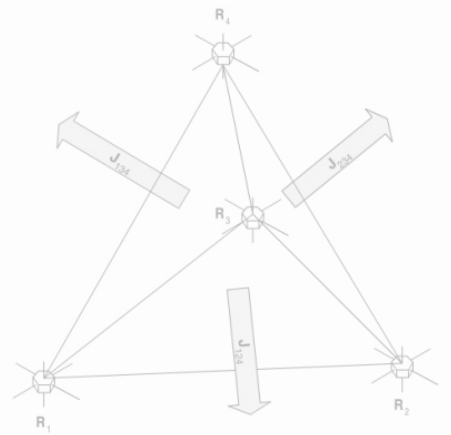


Synthetic Current Setup

True Current Configuration:

$$\mathbf{J}_0 = J_0 \frac{1}{\sqrt{3}}(1,1,1) = \frac{1}{\sqrt{3}}(1,1,1)\,\mathrm{A/m^2}$$

Uniform current flowing along space diagonal, with effective source at: $\mathbf{r}_0 = (0.4, 0.4, 0.4)$



Magnetic Field Calculation:

From that we generate the magnetic field at each spacecraft position using the Biot-Savart based forward model.

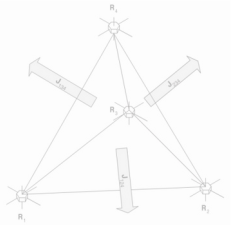
Synthetic Current Setup

Noise Model:

Real magnetometers have uncertainties. I add random directional noise:

$$\mathbf{B}_{\text{noisy}} = \mathbf{B}_{\text{true}} + \epsilon |\mathbf{B}_{\text{true}}| \hat{\mathbf{n}}$$

where $\epsilon \in \{5\%, 10\%, 15\%, 20\%, \ldots, 100\%\}$ and $\hat{\mathbf{n}}$ is a random unit vector.

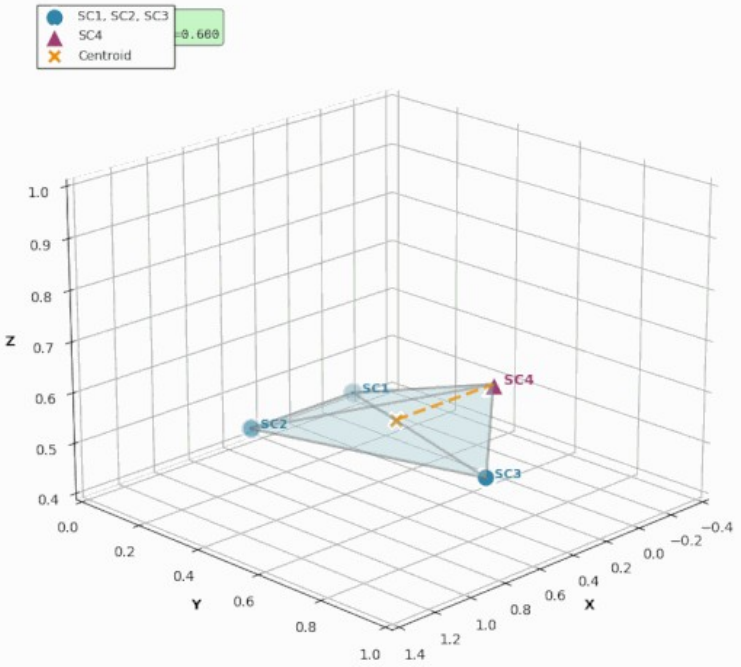


Performance Metric: Angular Error:

$$\theta = \arccos\left(\left|\frac{\mathbf{J}_0}{\|\mathbf{J}_0\|} \cdot \frac{\mathbf{J}_{\text{est}}}{\|\mathbf{J}_{\text{est}}\|}\right|\right)$$

In degrees. Small θ ($< 10^\circ$) → excellent reconstruction.

Geometry Scan Design



Geometric Quality Metrics

(Quantifying Mesh Constellation Quality)

Overview

To quantify constellation quality, we use standard measures from *Paschmann & Daly (1998):*

Volumetric Tensor Eigenanalysis – characterizes 3D configuration

Elongation and Planarity – measures geometric distortion

Combined Quality Factor – overall metric

Tetrahedron Geometric Factor (TQF) – volume-based assessment

Volumetric Tensor Eigenanalysis

The tensor R has three eigenvalues:

$$R^{(1)} \geq R^{(2)} \geq R^{(3)} \geq 0$$

These define semi-axes of an ellipsoid:

$$a = \sqrt{R^{(1)}}, \quad b = \sqrt{R^{(2)}}, \quad c = \sqrt{R^{(3)}}$$

Geometric Quality Metrics

(Quantifying Mesh Constellation Quality)

Elongation and Planarity:

$$E = 1 - \frac{b}{a}, P = 1 - \frac{c}{b}, Q = \sqrt{E^2 + P^2}$$

Elongation E: Measures cigar-like distortion
Planarity P: Measures pancake-like flattening

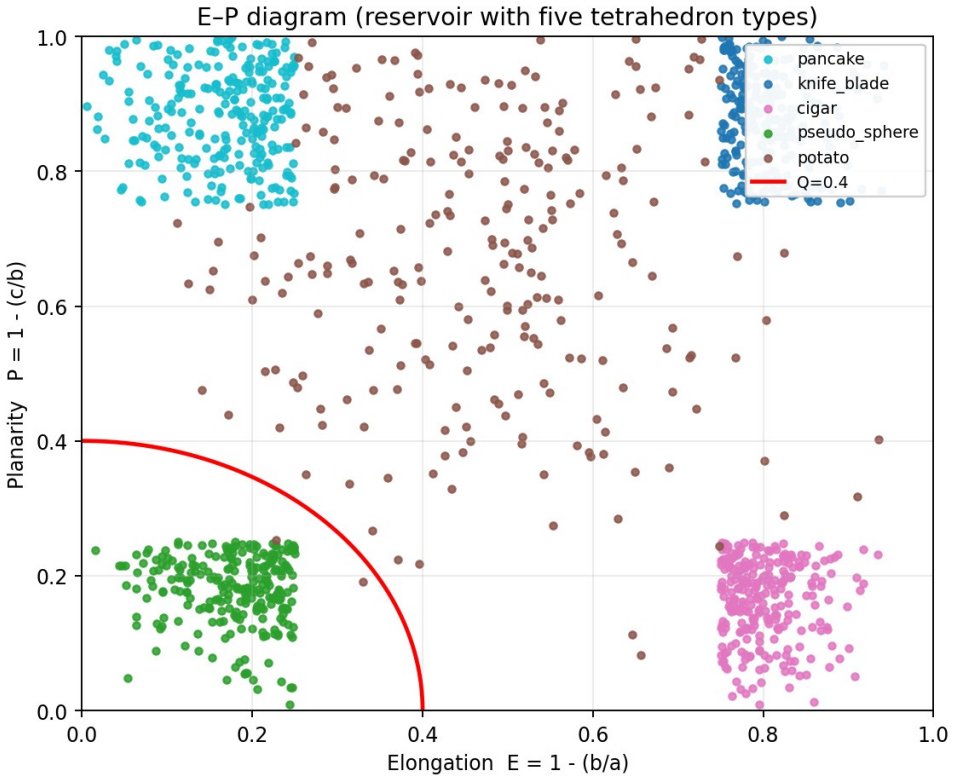
Tetrahedron Geometric Factor (TQF)

I compute the volume-based metric: $\mathrm{TQF} = \frac{V_{\text{actual}}}{V_{\text{regular}}},$

Where,

$$V_{\text{actual}} = \frac{1}{6}|(\mathbf{r}_4 - \mathbf{r}_1) \cdot [(\mathbf{r}_2 - \mathbf{r}_1) \times (\mathbf{r}_3 - \mathbf{r}_1)]|$$

$$V_{\text{regular}} = \frac{\sqrt{2}}{12} L^3 \quad (L = \text{average edge length})$$



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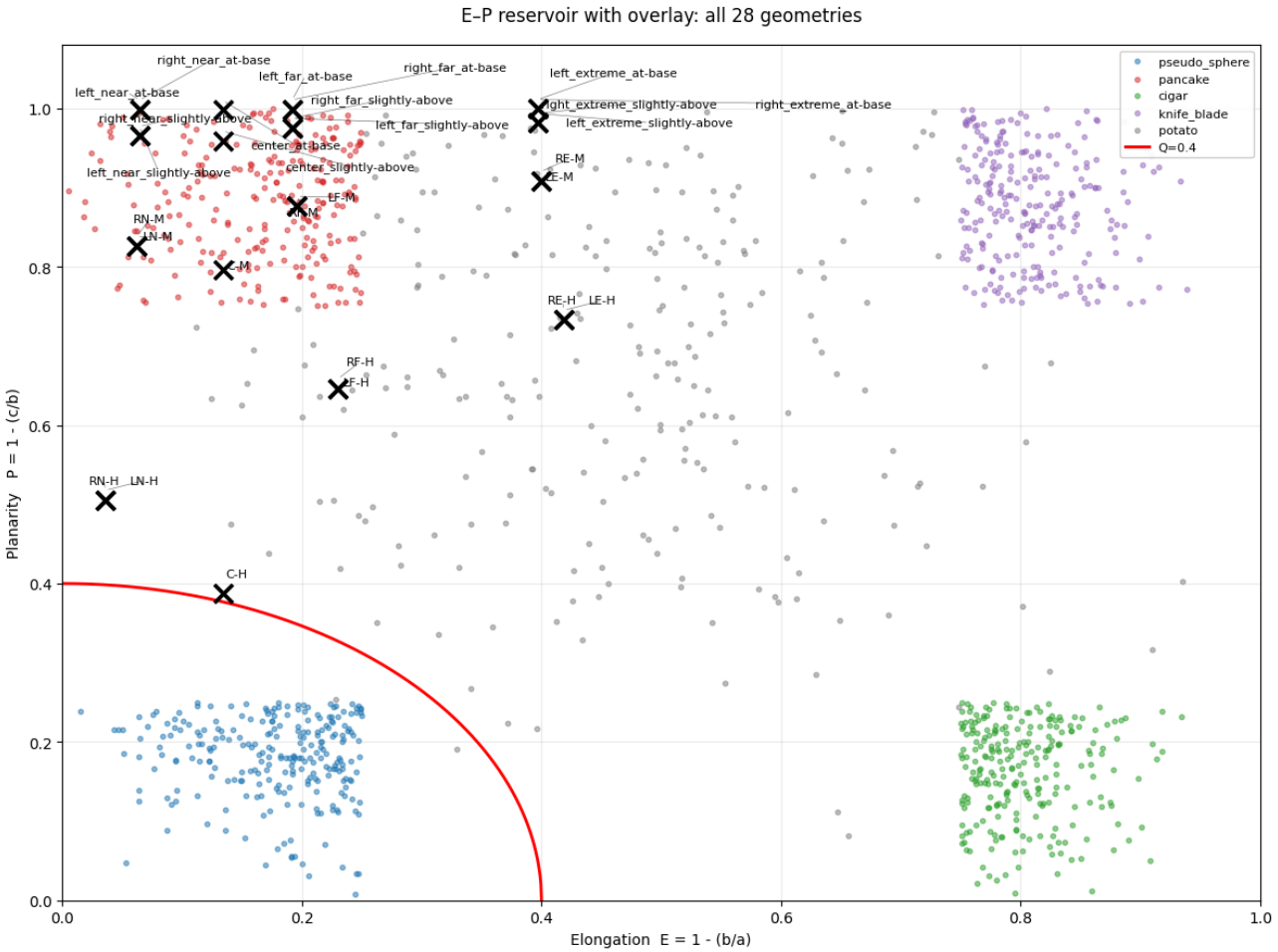
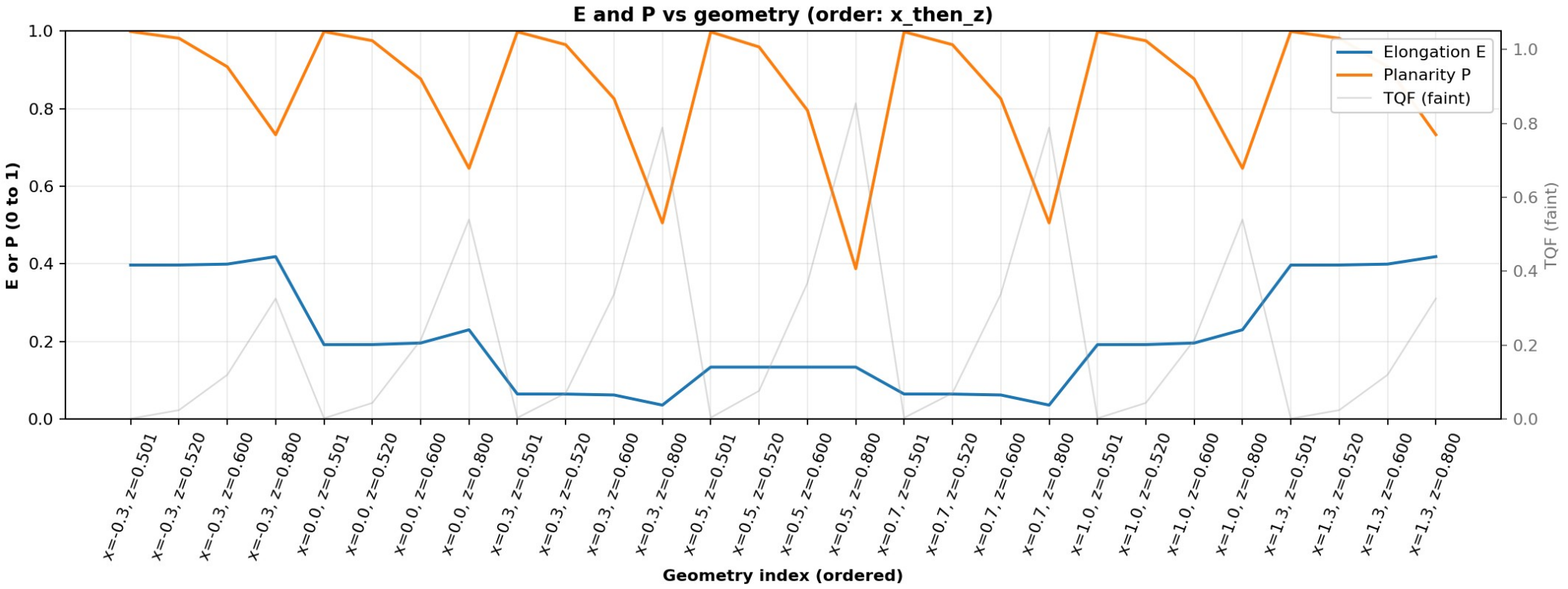


Geometry Scan Design



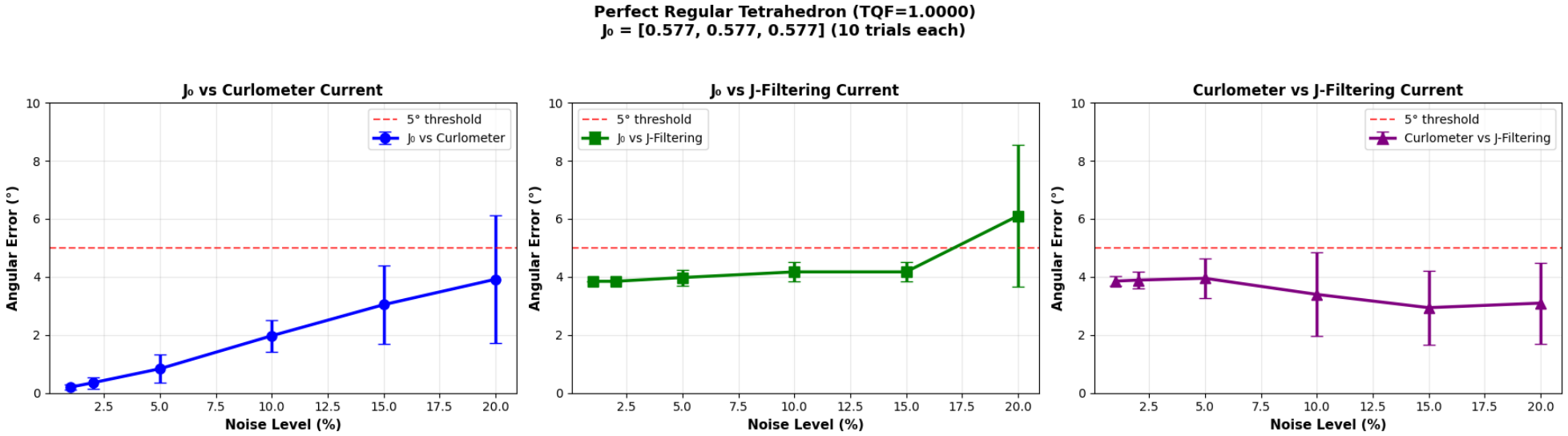
Geometry Scan Design

TQF Distribution

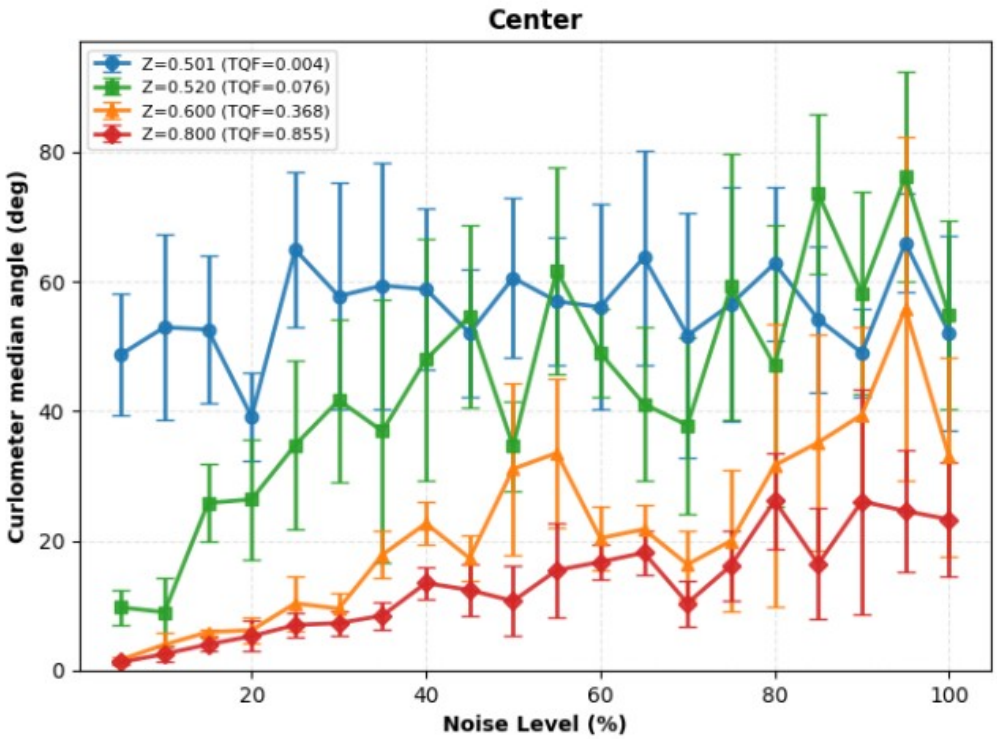
Best geometries (Center, Moderate-High Z): TQF ≈ 0.8–1.0

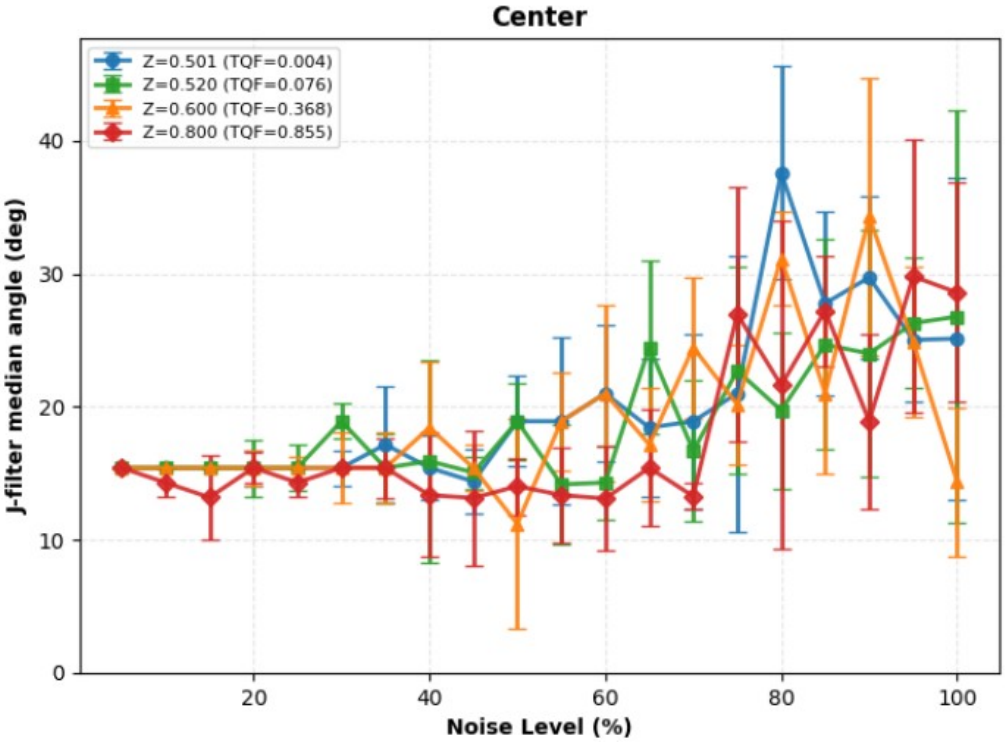
Degraded geometries (Extremes, Low Z): TQF < 0.3

Degenerate geometries (Extremes at Base): TQF < 0.1



Results – Curlometer and J- Filtering Performance





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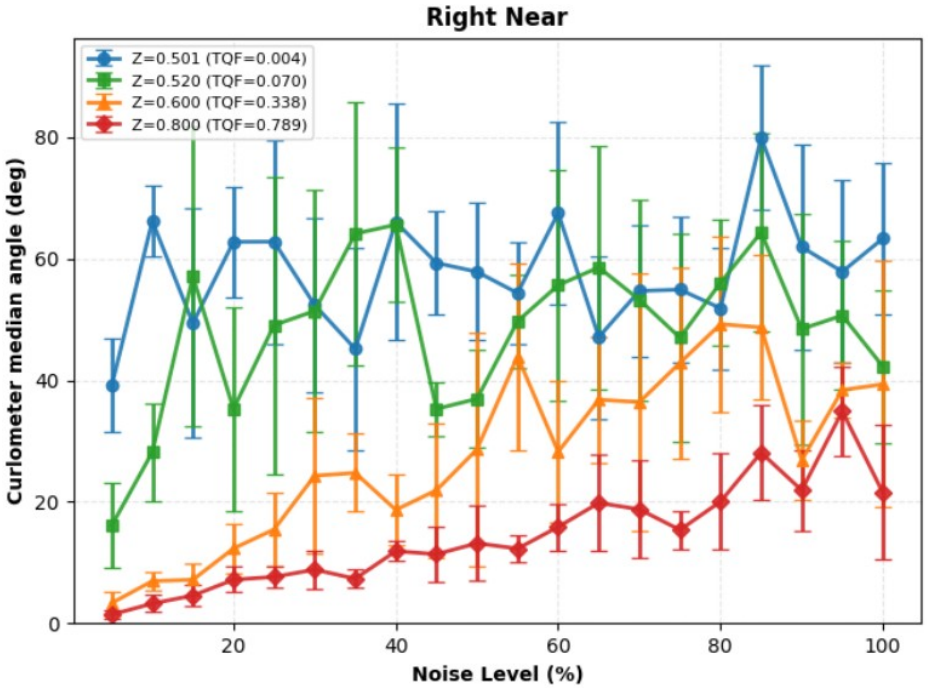

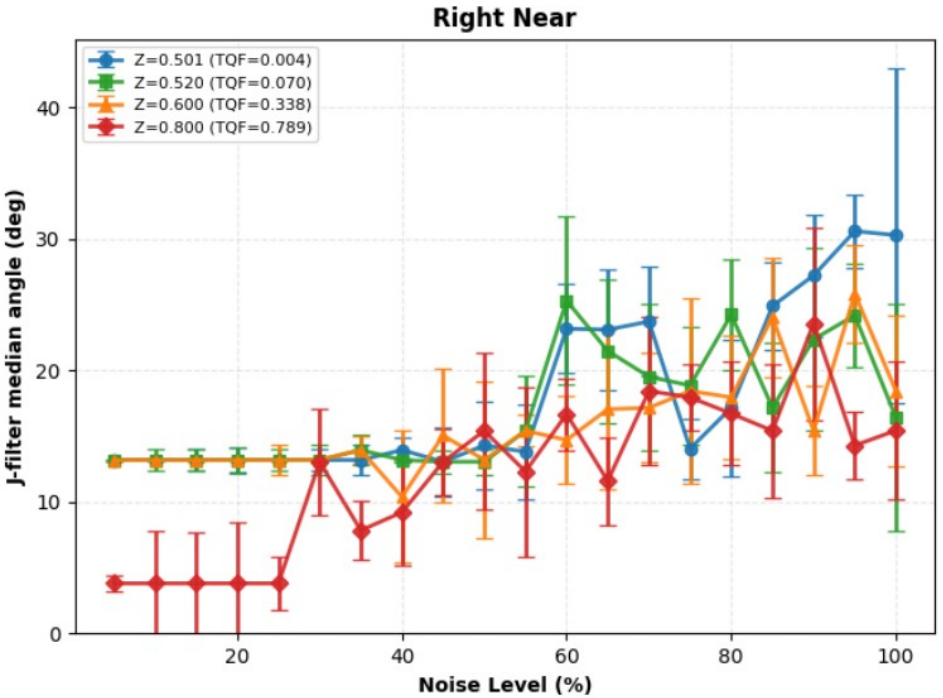

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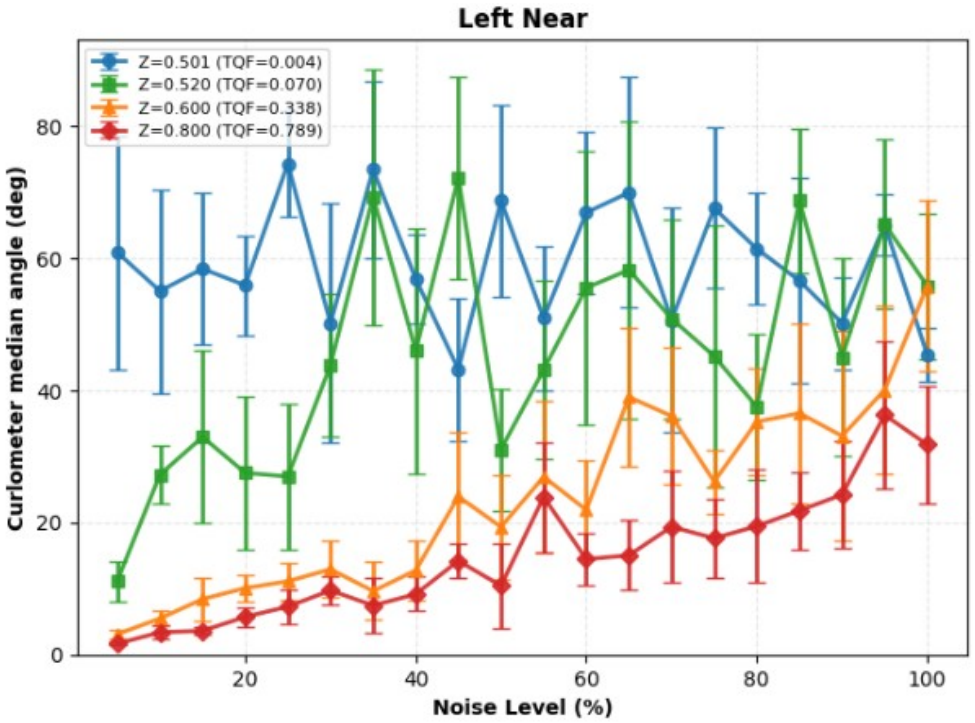

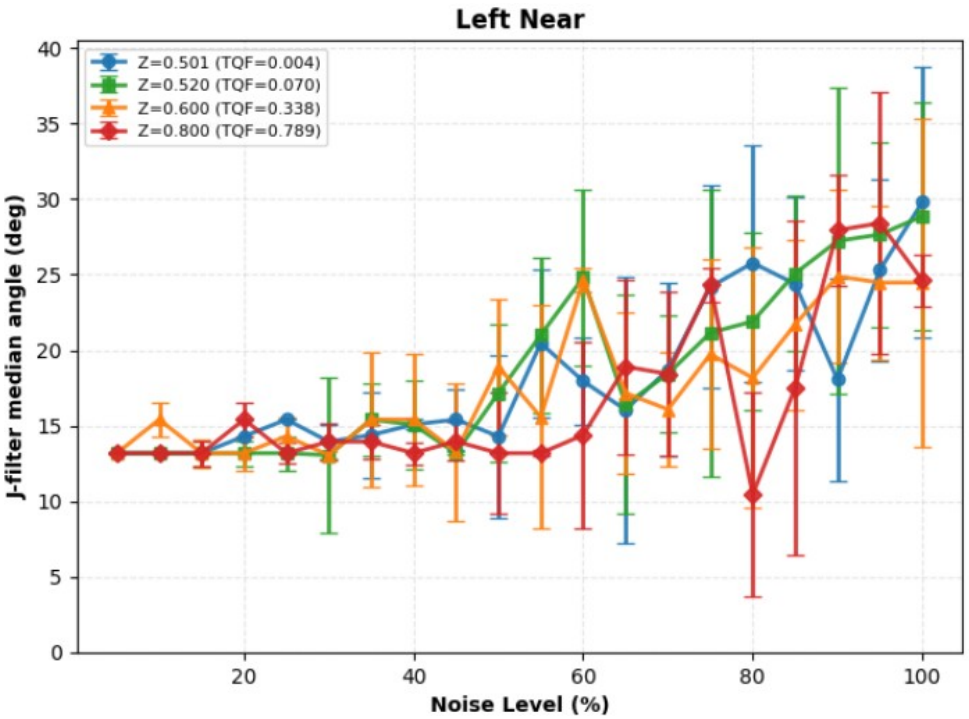

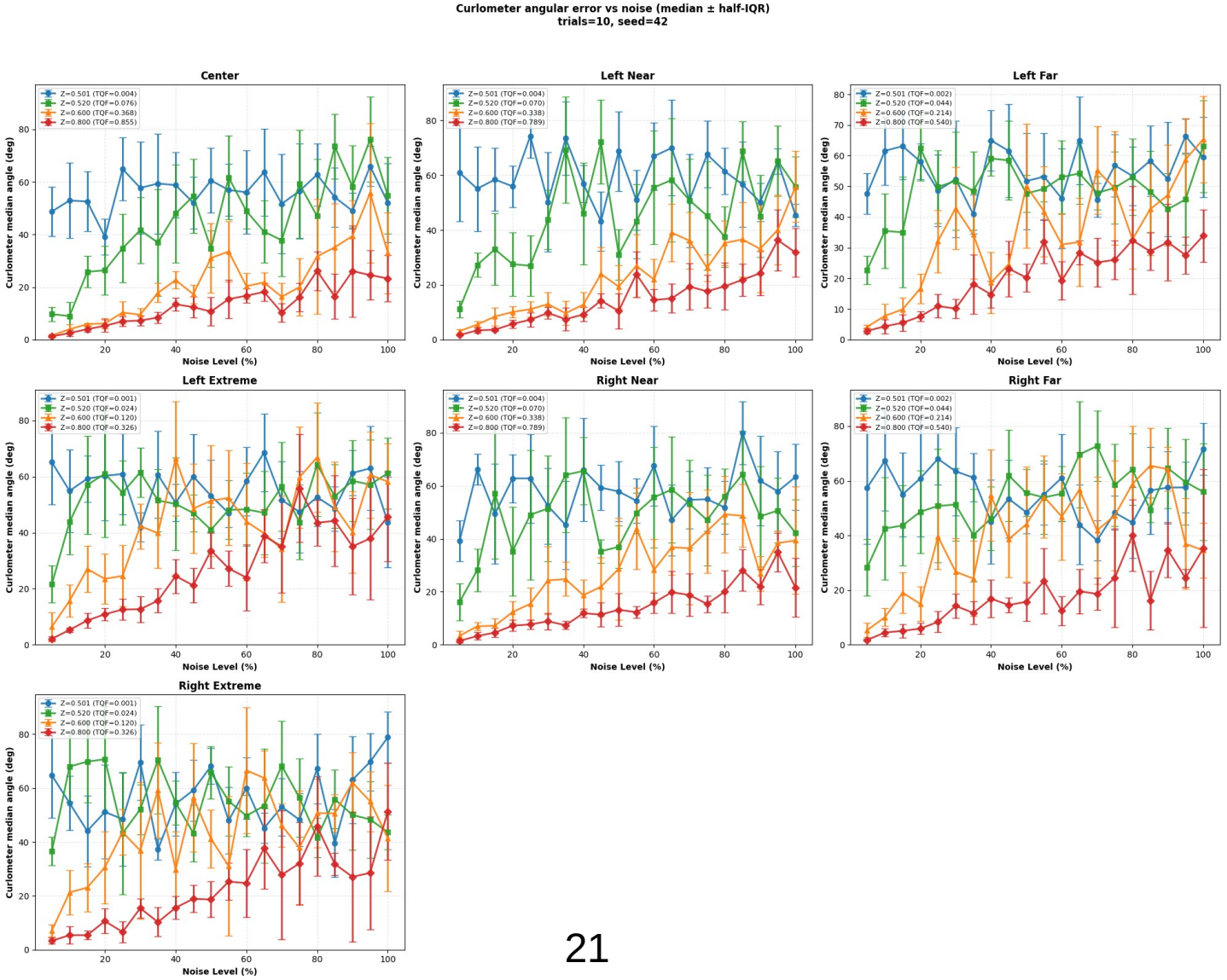
Curlometer angular error vs noise (median ± half-IQR)
trials=10, seed=42
Center
Z=0.501 (TQF=0.004)
Z=0.520 (TQF=0.076)
Z=0.600 (TQF=0.368)
Z=0.800 (TQF=0.855)
Left Near
Z=0.501 (TQF=0.004)
Z=0.520 (TQF=0.070)
Z=0.600 (TQF=0.338)
Z=0.800 (TQF=0.789)
Left Far
Z=0.501 (TQF=0.002)
Z=0.520 (TQF=0.044)
Z=0.600 (TQF=0.214)
Z=0.800 (TQF=0.540)
Left Extreme
Z=0.501 (TQF=0.001)
Z=0.520 (TQF=0.024)
Z=0.600 (TQF=0.120)
Z=0.800 (TQF=0.326)
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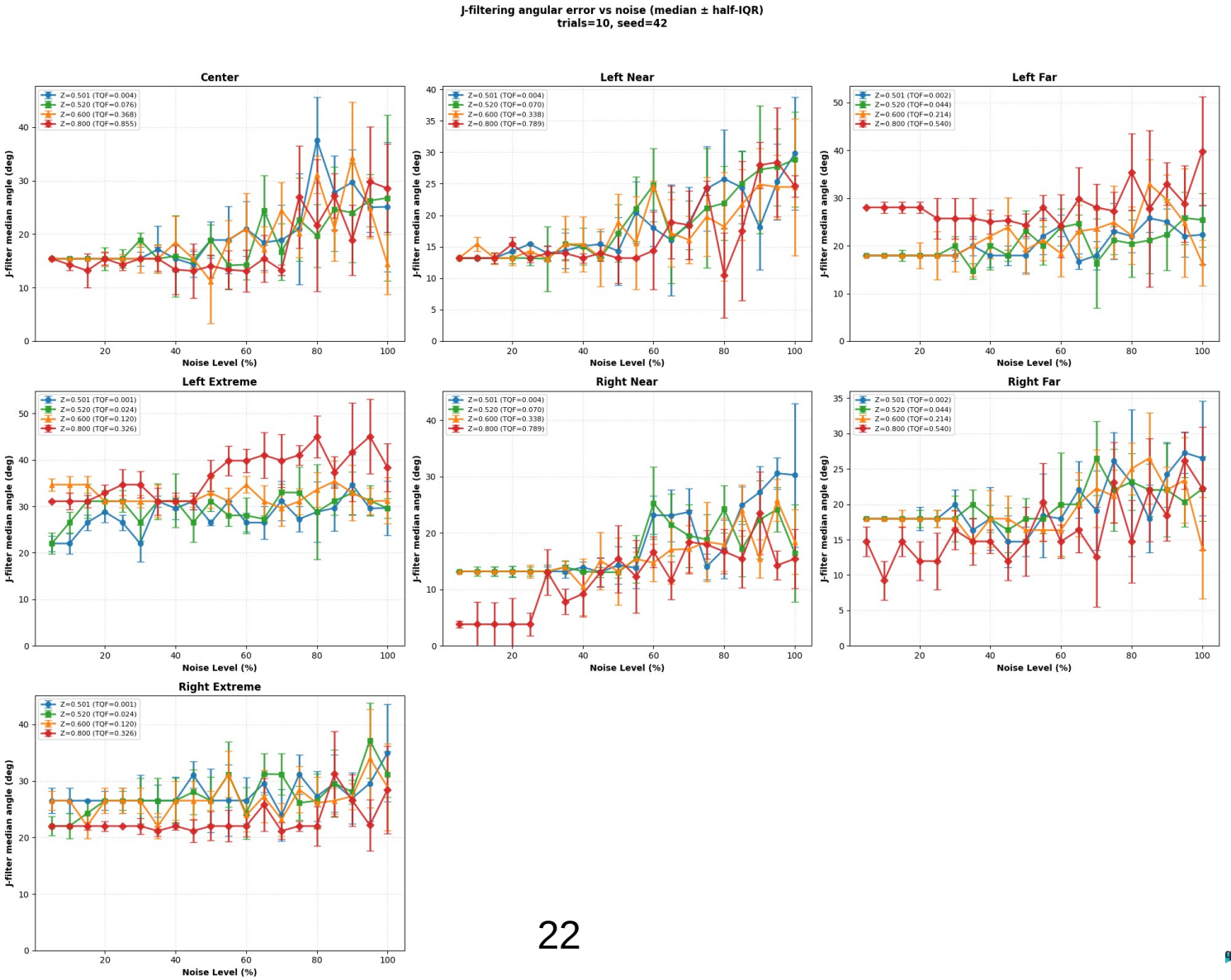
Z=0.520 (TQF=0.024)
Z=0.600 (TQF=0.120)
Z=0.800 (TQF=0.326)
Curlometer median angle (deg)
Noise Level (%)


J-filtering angular error vs noise (median ± half-IQR)
trials=10, seed=42
Center
Z=0.501 (TQF=0.004)
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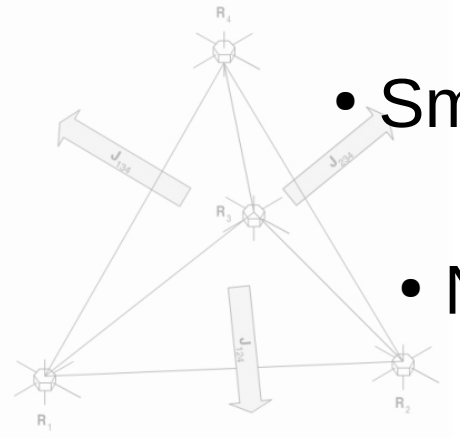
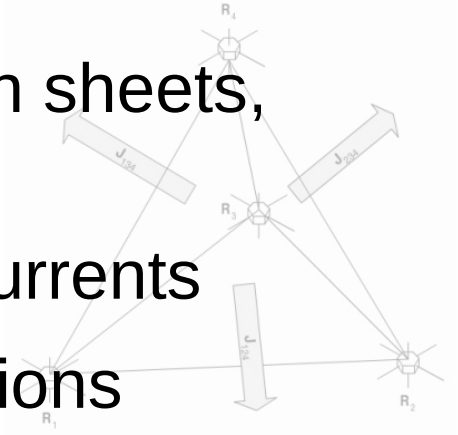
Z=0.600 (TQF=0.120)
Z=0.800 (TQF=0.326)
J-filter median angle (deg)
Noise Level (%)

Conclusions and Future Work



- Small-scale current structures (thin sheets, gradients)
- Non-uniform and time-varying currents
- 5 and 6-spacecraft configurations
- Real MMS data validation



Acknowledgments & References

Thank you for your attention.

Key References:

- *Chakraborty, M., Kretzschmar, M., & Pinçon, J.L. (2024). J-Filtering: A new method to determine current distribution from multi-points magnetic field measurements. JGR:*
- *Space Physics (submitted).*
- *Dunlop, M.W., et al. (2021). Curlometer technique and applications. JGR: Space Physics, 126(11), e29538.*
- *Paschmann, G., & Daly, P.W. (1998). Analysis Methods for Multi-Spacecraft Data. ISSI Scientific Reports Series SR-001.*
- *Pinçon, J.L., & Lefeuvre, F. (1991). Local characterization of homogeneous turbulence in a space plasma from simultaneous measurements of field components at several points in space. JGR, 96.*

D Monte Carlo flowchart

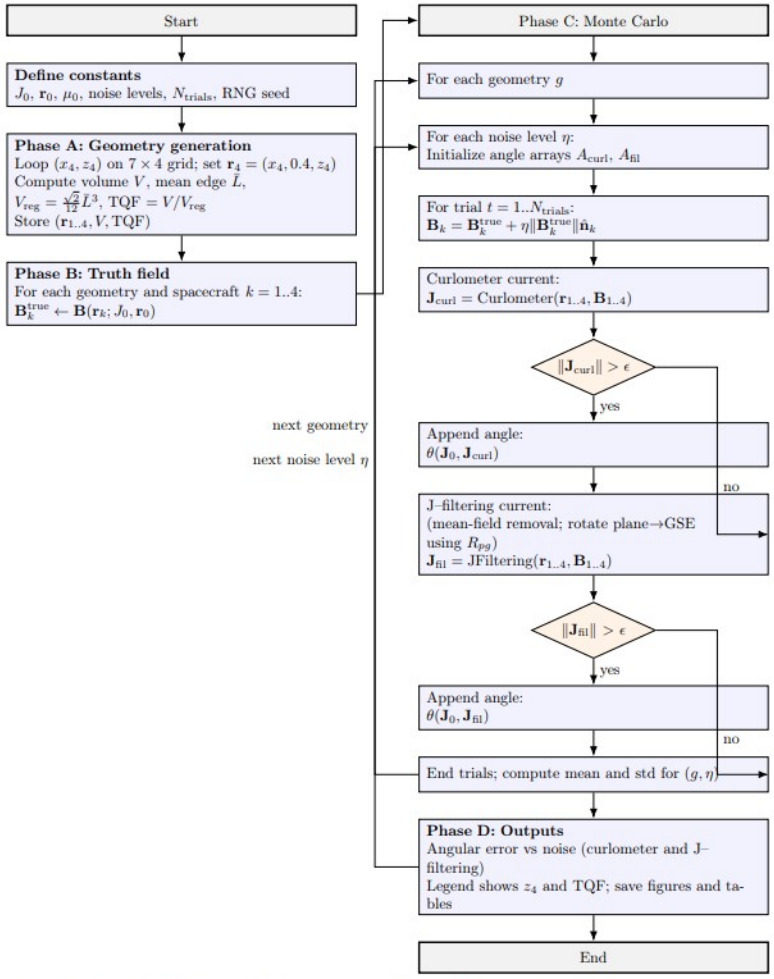


Figure 8: Two-lane flowchart of the 28-geometry Monte Carlo analysis pipeline for angular current estimation.